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FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

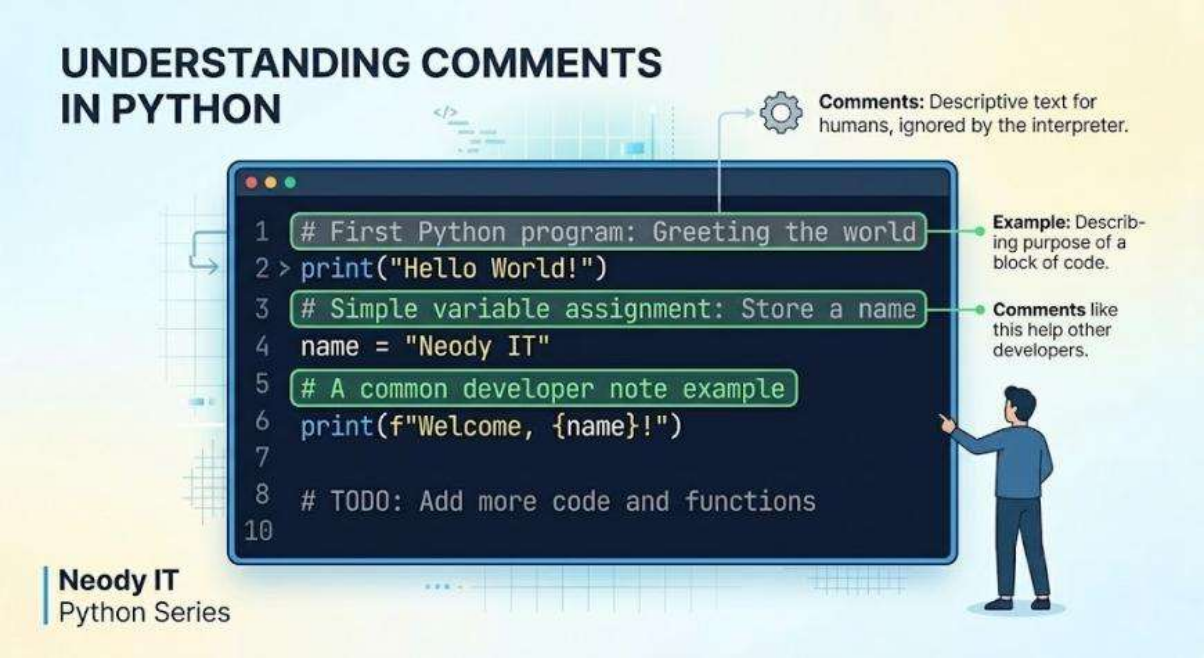
Classes : Offline & Online

Daily Teaching : 2 Hours

Total Classes : 100

Level 2 – Young Technologist (7th–8th Standard)

Lesson 19: Comments in Python



Learning Objectives

By the end of this lesson, students will be able to:

- ✅ Understand what comments are in Python.
- ✅ Learn why comments are important.
- ✅ Write single-line and multi-line comments.
- ✅ Add comments to make programs easier to understand.
- ✅ Follow good coding practices.

What are Comments?

A **comment** is a note written inside a program to explain the code.

- Comments are **ignored by Python**.
- They are written to help **students and programmers** understand the program.
- Comments make programs easier to read and maintain.

Real-Life Example

Imagine you have a school notebook.

You write:

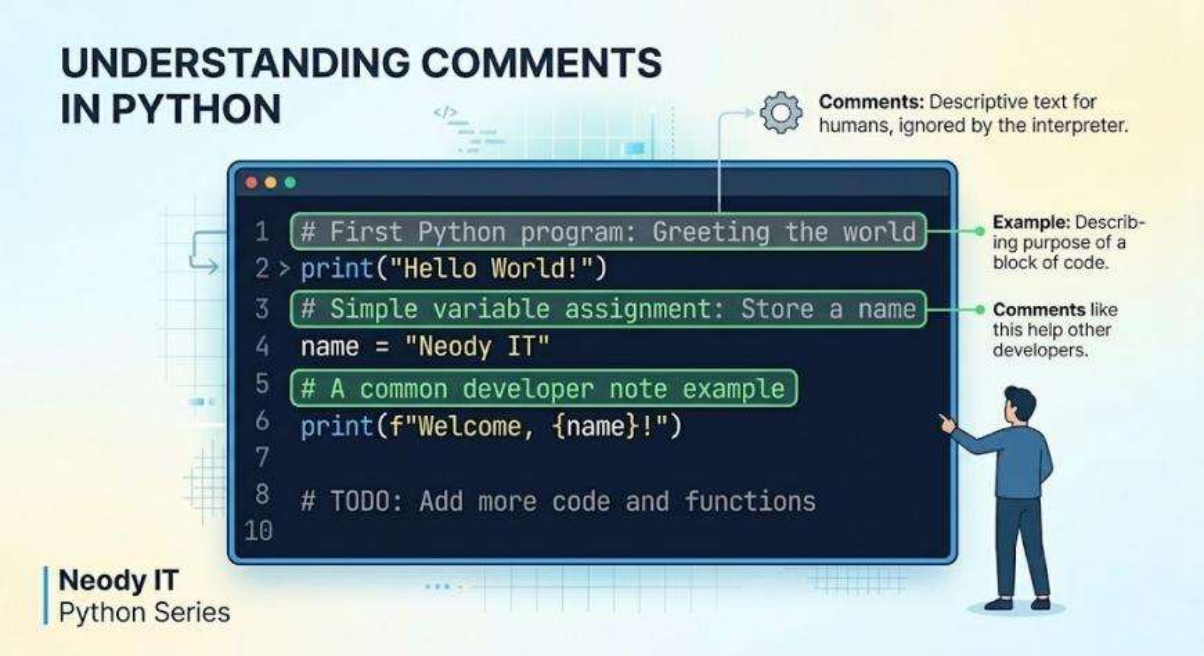
- **Lesson Title**
- **Important Notes**
- **Headings**

These notes help you understand the lesson later.

Similarly, **comments help programmers understand the code**.

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What are Comments?

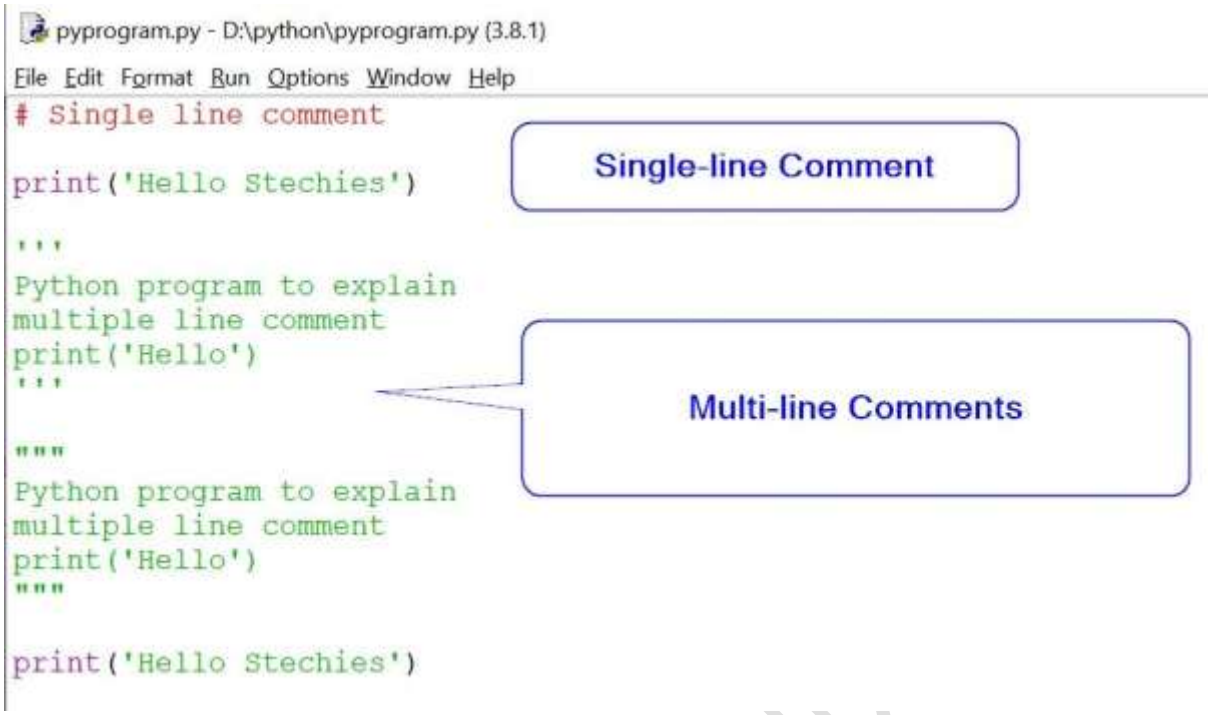


main.py

```python
# initializing two variables
a = 5
b = 7

# adding a and b
result = a + b

# printing the result of the addition
print(result)
```



Why Do We Use Comments?

Comments help us to:

- 📖 Explain the purpose of the program.
- 👩‍🏫 Help teachers and friends understand the code.
- 🐞 Find and fix mistakes more easily.
- 📝 Remember what each part of the program does.
- 🤝 Work with other programmers.

Types of Comments in Python

There are **two main types** of comments:

Type	Symbol	Example
Single-Line Comment	#	# This is a comment

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Type	Symbol	Example
Multi-Line Comment	''' ''' or """ """	"""Multiple lines"""

1️⃣ Single-Line Comments

A **single-line comment** starts with the **#** symbol.

Everything after # on that line is ignored by Python.

Example

```
# This program prints a message


print("Hello Students")
```

Output

```
Hello Students
```

Another Example

```
# Displaying welcome message


print("Welcome to Python")
```

Output

```
Welcome to Python
```

Single-Line Comment Example

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main.py

```
# initializing two variables
a = 5
b = 7

# adding a and b
result = a + b

# printing the result of the addition
print(result)

```

Creating a Comment

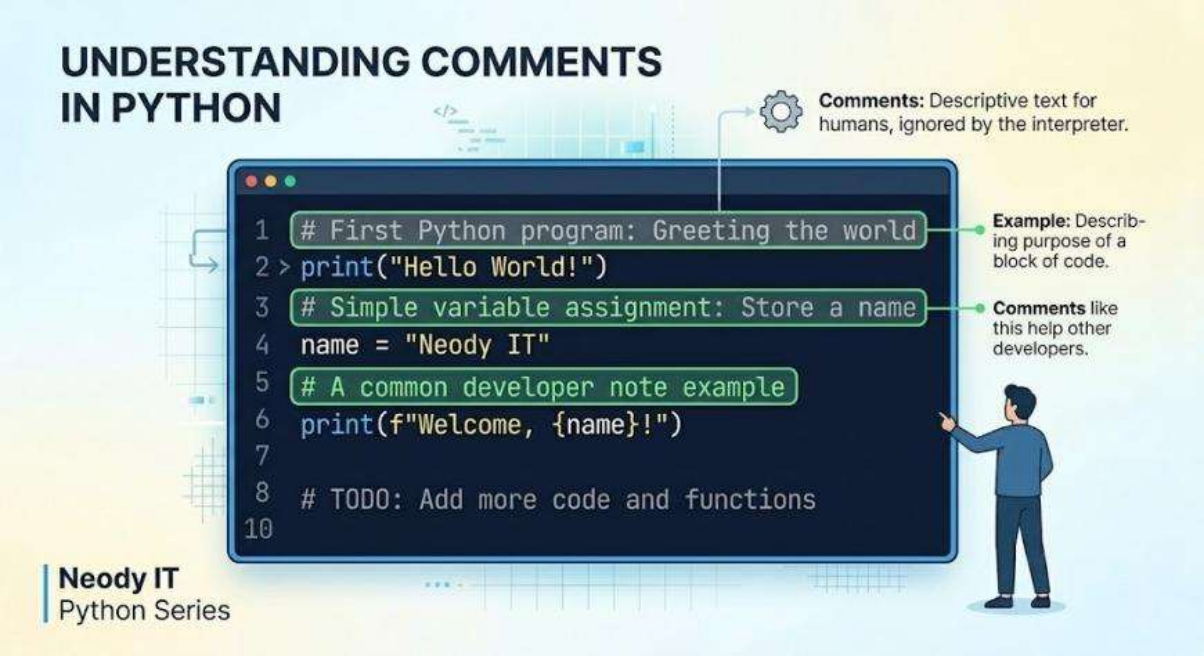
Comments starts with a #, and Python will ignore them:

Example

```
#This is a comment
print("Hello, World!")
```

Comments can be placed at the end of a line, and Python will ignore the rest of the line:

```
print("Hello, World!") #This is a comment
```



2️⃣ Multi-Line Comments

Sometimes we want to write **more than one line** of explanation.

Python programmers often use **triple quotes** (''' ''' or """ """) for long descriptions.

Example

"""

This program

prints a welcome

message.

"""

print("Welcome")

Output

Welcome

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Note: Triple-quoted text is technically a string literal. It is commonly used like a multi-line comment when it is not assigned to a variable.

Comments Before Code

```
# Store student name

name = "Rahul"

# Display student name

print(name)
```

Output

Rahul

Comments Beside Code

```
age = 12      # Student age

print(age)
```

Output

12

Comments in Calculations

```
# Addition of two numbers
```

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```
a = 10

b = 20


print(a + b)
```

Output

30

Comments in Input Programs

```
# Ask user for name


name = input("Enter your name: ")


# Display greeting


print("Hello", name)
```

Comments in Math Programs

```
# Find area of rectangle


length = 8

breadth = 5


area = length * breadth


print(area)
```

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Output

40

Good Comment Example

```
# Calculate total marks

math = 80
science = 75

total = math + science

print(total)
```

Poor Comment Example

```
# Number

a = 5

# Number

b = 10

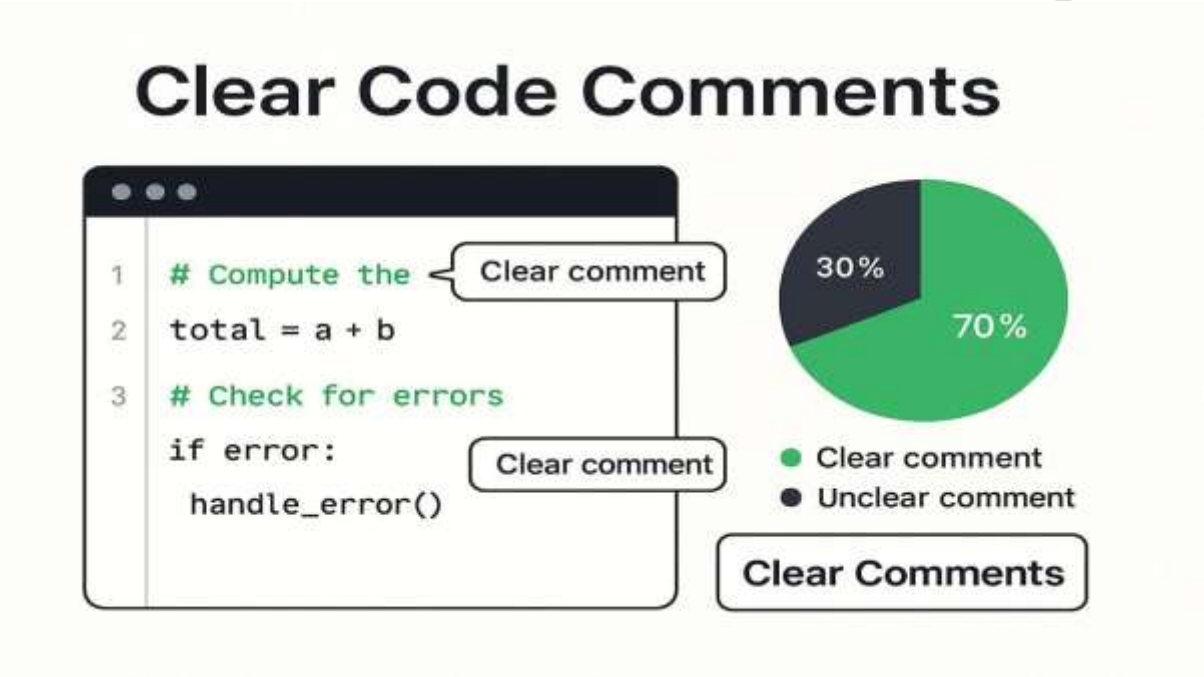
# Add
```

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print(a + b)

👉 Better comments explain **why** or **what** the code does, not just repeat the code.

Good vs Poor Comments





Practical Session

Practical Activity 1

Print Your Name with Comments

```
# Display student name


print("Rahul")
```

Practical Activity 2

Greeting Program

```
# Ask for student name

name = input("Enter your name: ")


# Print greeting


print("Welcome", name)
```

Practical Activity 3

Addition Program

```
# Enter two numbers

a = int(input("First Number: "))
b = int(input("Second Number: "))
```

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```
# Calculate sum

print(a + b)
```

Practical Activity 4

Area of Rectangle

```
# Input length and breadth

length = int(input("Length: "))
breadth = int(input("Breadth: "))

# Calculate area

area = length * breadth

# Display result

print("Area =", area)
```

Practical Activity 5

Multiplication Table

```
# Print multiplication table of 5
```

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```
for i in range(1, 11):
    print("5 x", i, "=", 5 * i)
```

Practical Activity 6

Student Details

```
# Get student information


name = input("Name: ")
age = input("Age: ")
city = input("City: ")


# Display information


print(name)
print(age)
print(city)
```

Classroom Activities

Activity 1

Add comments to this program:

```
a = 5
b = 10


print(a + b)
```

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Activity 2

Write a program to print your school name with comments.

Activity 3

Write a program to calculate the product of two numbers and explain each step with comments.

Activity 4

Read a friend's program and explain what each comment means.

Real-Life Applications

Situation	How Comments Help
School Project	Explain each step
Team Coding	Help classmates understand the code
Homework	Make programs easier to revise
Future Editing	Remember how the program works
Debugging	Find errors more quickly

Best Practices for Writing Comments

✅ Write clear and short comments.

✅ Explain **why** the code is written.

✅ Keep comments up to date if the code changes.

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✅ Use comments before important sections.

❌ Do not write unnecessary comments for every single line.

Common Mistakes

❌ **Forgetting #**

Wrong

```
This is a comment


print("Hello")
```

Python will show an error.

✅ **Correct**

```
# This is a comment


print("Hello")
```

❌ **Writing misleading comments**

```
# Add two numbers

print(10 - 5)
```

The comment says **add**, but the code **subtracts**.

✅ **Correct**

```
# Subtract two numbers
```

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```
print(10 - 5)
```

Keywords

- **Comment** – A note that explains the code.
- **Single-Line Comment** – A comment beginning with #.
- **Multi-Line Comment** – A description written using triple quotes (''' ''' or """ """).
- **Documentation** – Written information that explains a program.
- **Readability** – How easy a program is to understand.

Lesson Summary

- Comments explain programs.
- Python ignores comments while running the code.
- Use # for single-line comments.
- Triple quotes can be used for longer descriptions.
- Good comments make programs easier to read, understand, and maintain.
- Comments are helpful for students, teachers, and programmers working together.

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Quiz

Multiple Choice Questions

1. Which symbol is used for a single-line comment?

A) @

B) #

C) &

D) $

✅ **Answer: B**

2. Does Python execute comments?

A) Yes

B) No

✅ **Answer: B**

3. Why do we write comments?

A) To decorate the program

B) To explain the code

C) To delete variables

D) To increase computer speed

✅ **Answer: B**

4. Which of the following is a correct comment?

A)

Print Hello

B)

Print Hello

C)

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// Print Hello

D)

<!-- Print Hello -->

✅ **Answer: A**

5. Which comment is more useful?

A)

Variable

B)

Calculate total marks of the student

✅ **Answer: B**

Homework

1. Write a program to print your name and add comments.
2. Write a program to calculate the sum of two numbers with comments.
3. Write a greeting program using comments.
4. Write a program to find the area of a rectangle and explain each step using comments.
5. Ask your friend to read your program and check whether your comments are easy to understand.

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